

Hadoop and MapReduce Setup - Step-by-Step Breakdown

1. Hadoop Services Management (Start HDFS and YARN)

Start HDFS (Distributed File System):

Command: start-dfs.sh

Error: Daemons already running. Stop first and clear .pid files if needed.

Tip: To restart cleanly, use:

stop-dfs.sh

stop-yarn.sh

rm /tmp/hadoop-student-*.pid

start-dfs.sh

start-yarn.sh

Start YARN (Yet Another Resource Negotiator):

Command: start-yarn.sh

Error: ResourceManager and NodeManager already running.

Check Running Hadoop Services:

Command: jps

Output:

- SecondaryNameNode
- NameNode
- ResourceManager
- DataNode
- NodeManager
- Jps

2. MapReduce Setup Using Python Scripts

Write Mapper and Reducer Scripts:

Commands:

nano mapper.py

nano reducer.py

Make Scripts Executable:

Commands:

```
chmod +x mapper.py
```

```
chmod +x reducer.py
```

3. Testing MapReduce Locally (Without Hadoop Streaming Yet)

Create a Sample Input File:

Command:

```
echo "hello world welcome to the world of big data hello" > input.txt
```

Run MapReduce Pipeline Manually:

Command:

```
cat input.txt | ./mapper.py | sort | ./reducer.py
```

Expected Output:

big 1

data 1

hello 2

of 1

the 1

to 1

welcome 1

world 2

Explanation:

1. mapper.py -> Outputs key-value pairs like (word, 1)
2. sort -> Groups same words together
3. reducer.py -> Sums counts for each word